HVAC System

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Abstract

Evaporative coolers, also known as swamp coolers, are very effective and cost efficient according to grandma and grandpa. It is June 8 and 106°F outside, but it is 'freezing' in their house. Interestingly enough, not only is it inexpensive to run a swamp cooler in the summer, but it roughly averages $10 a month. Nonetheless, evaporative coolers are not as commonly used as one would imagine if they were as 'great' as they seem to be. To solve the mystery of why evaporative coolers are no longer used for most houses in Phoenix, many factors have to be accounted for, including energy usage of different cooling devices, cost analysis based on the energy consumption, climate throughout the year. Considering these factors, an HVAC system for an average sized home of 1832-sq. ft. is designed.

HVAC System

History and Diagrams

The evaporative cooler was first created in Phoenix, Az in 1908 by Oscar Palmer (Evaporative coolers, 2019). A few years later, two professors from the University of Arizona, Martin Thornburg and Paul Thornburg, created instructions so others would be able to make their own evaporative coolers (Evaporative coolers, 2019). By 1936, thousands of homes across Arizona had their own evaporative coolers after the production began (Evaporative coolers, 2019), and Arizona became the largest manufacturer of evaporative coolers at the same time manufacturing was spreading across the Southwest (Potts, 2005). These evaporative coolers soon became known as “swamp coolers”, named after the algae that would develop inside of them (Historic vehicle association, 2013). However, once air conditioners appeared in the 1950s, the majority of people abandoned their evaporative coolers for the new AC units, which seems to have stuck around to this day (Potts, 2005). Evaporative coolers are not in use today as much as they once were because any energy savings they do provide is minimal. Also, they tend to be noisy and faulty, especially if they are older units. In addition, evaporative coolers do not work well in hot and humid weather, which is nearly guaranteed for a period of time throughout the year in Phoenix during monsoon season. Therefore, it is easier for homeowners to simply just use an air conditioning unit for year-round cooling instead.

Evaporative Cooling System

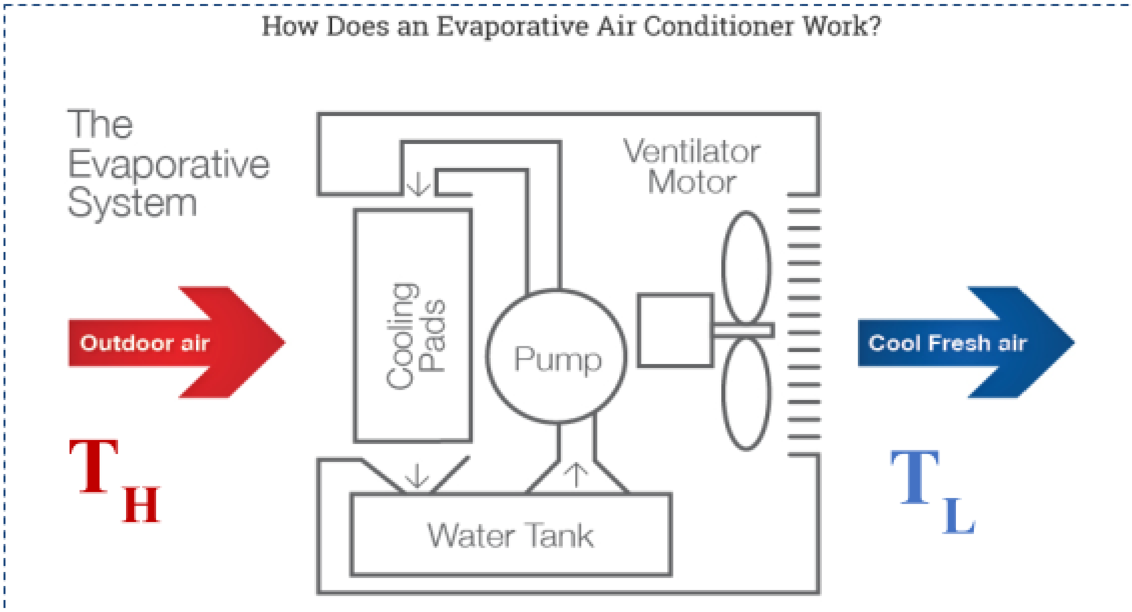

Figure 1: Evaporative Cooler

(Evaporative cooling, 2019)

$$\dot{E}_{in} - \dot{E}_{out} = 0$$

$$\dot{m}_{vapor,T_H} h_{vapor,T_H} + \dot{m}_{H_2O,T_L} h_{H_2O,T_L} + \dot{W}_{in} = \dot{m}_{vapor,T_H}\left(h_{vapor,T_H} + Xh_{vapor,T_L}\right)$$

Warm air from the outdoors is drawn into the evaporation system by a fan driven by a motor. As the warm air goes past the cooling pads, thermal energy from the air transfers over to the water inside the pads. The heated water then evaporates and turns into vapor that mixes with the output air, resulting in a decrease in temperature. That is why the relative humidity of the air inside is higher than the air outside, and that air circulation is crucial when using an evaporative cooler. The higher the relative humidity it is on the inside, the less efficient the evaporative cooler is because there is a limit to how much water the air can hold. The ideal environmental relative humidity ranges between 0-30% (Dieter, n.d.). Thus, evaporative coolers are not suitable to many places.

Heat Pump

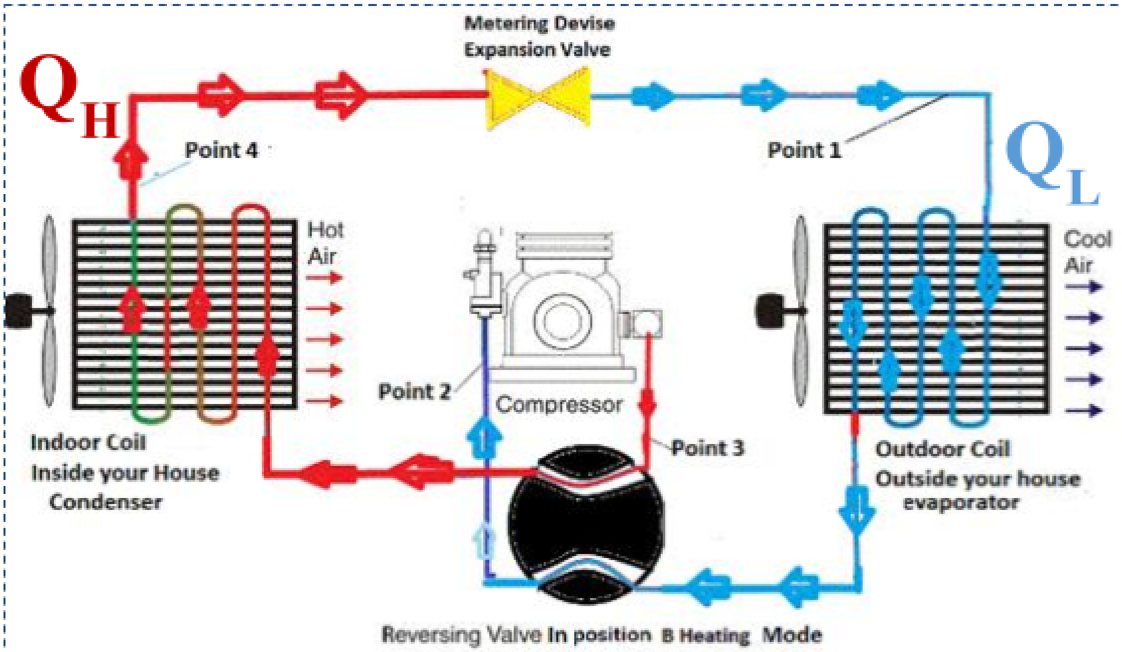

Figure 2: Heat Pump

(Heating and cooling, inc., 2019)

$$\dot{E}_{in} - \dot{E}_{out} = 0$$

$$\dot{Q}_{L} + \dot{W}_{in} = \dot{Q}_{H}$$

Refrigerant is the working fluid in a heat pump. The expansion valve is the first place the refrigerant passes through. Here, the refrigerant is converted to a low-pressure saturated mixture. In this state, the refrigerant then enters the outdoor coil which acts as the evaporator. This is where the refrigerant absorbs the heat from the outside air and becomes a low-temperature vapor. The vaporized refrigerant passes through an accumulator before entering the compressor because the accumulator collects any remaining fluid refrigerant, that way, the vaporized refrigerant can be fully compressed by the compressor, which causes it to heat up before entering the indoor coil, which acts as the condenser. The heated-up refrigerant vapor thus transfers its heat to the indoor space as it passes through the indoor coil before returning to the expansion valve as a liquid again after losing its heat (Natural resources Canada, 2017).

Conventional Cooling System

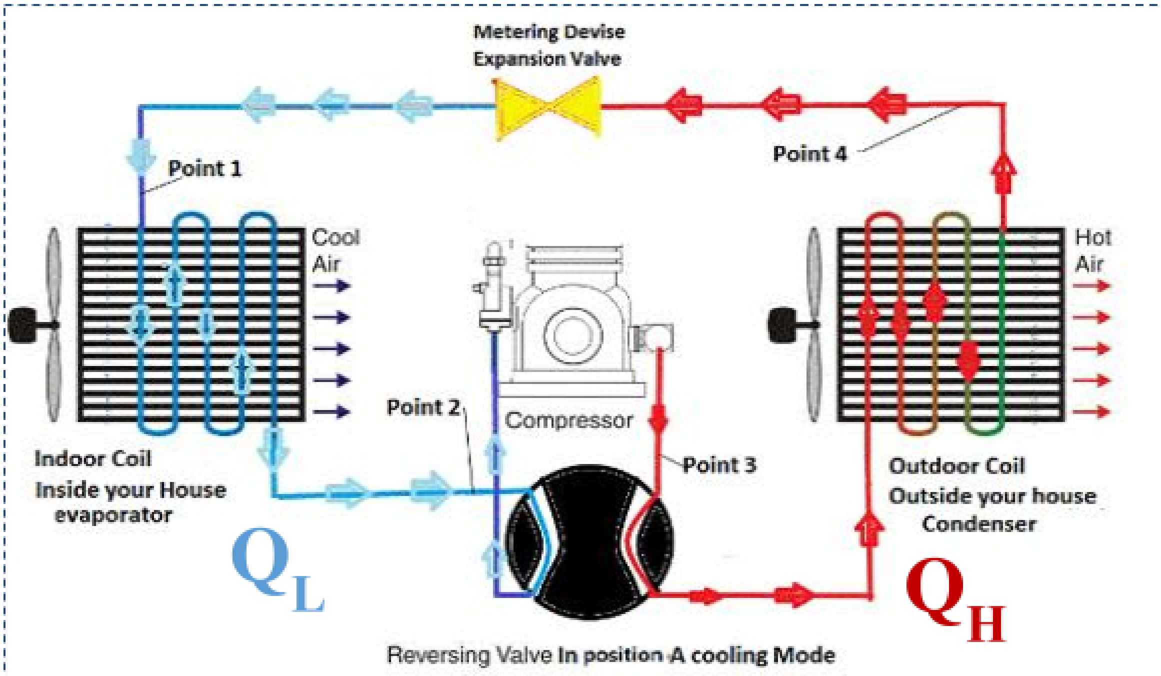

Figure 3: Air Conditioning

(Heating and cooling, inc., 2019)

$$\dot{E}_{in} - \dot{E}_{out} = 0$$

$$\dot{Q}_H + \dot{W}_{in} = \dot{Q}_L$$

A conventional cooling system (a/c) Is the same as a heat pump except that it operates in the reverse. This is why some HVAC systems are simply a heat pump with a reversing valve: the reversing valve allows the heat pump to either pull heat from the indoors to the outdoors and vice versa. Working as an a/c system, first, the liquid refrigerant enters the expansion valve and changes state into a low-pressure saturated mixture, the state in which it enters the indoor coil, which acts as the evaporator. In here, the refrigerant absorbs the indoor heat and becomes a low-temperature vapor. This vapor then passes through the accumulator, which accumulates excess liquid refrigerant so that the rest of the refrigerant is fully vaporized as it enters the compressor, which then compresses the refrigerant and causes it to heat up into a gas. Lastly, the refrigerant leaves the compressor and enters the outdoor coil, which acts as the condenser, and the heat from the refrigerant is passed to the outside air, causing the refrigerant to return to its liquid state before returning to the expansion valve and beginning the process again (Natural resources Canada, 2017).

Electric Heater

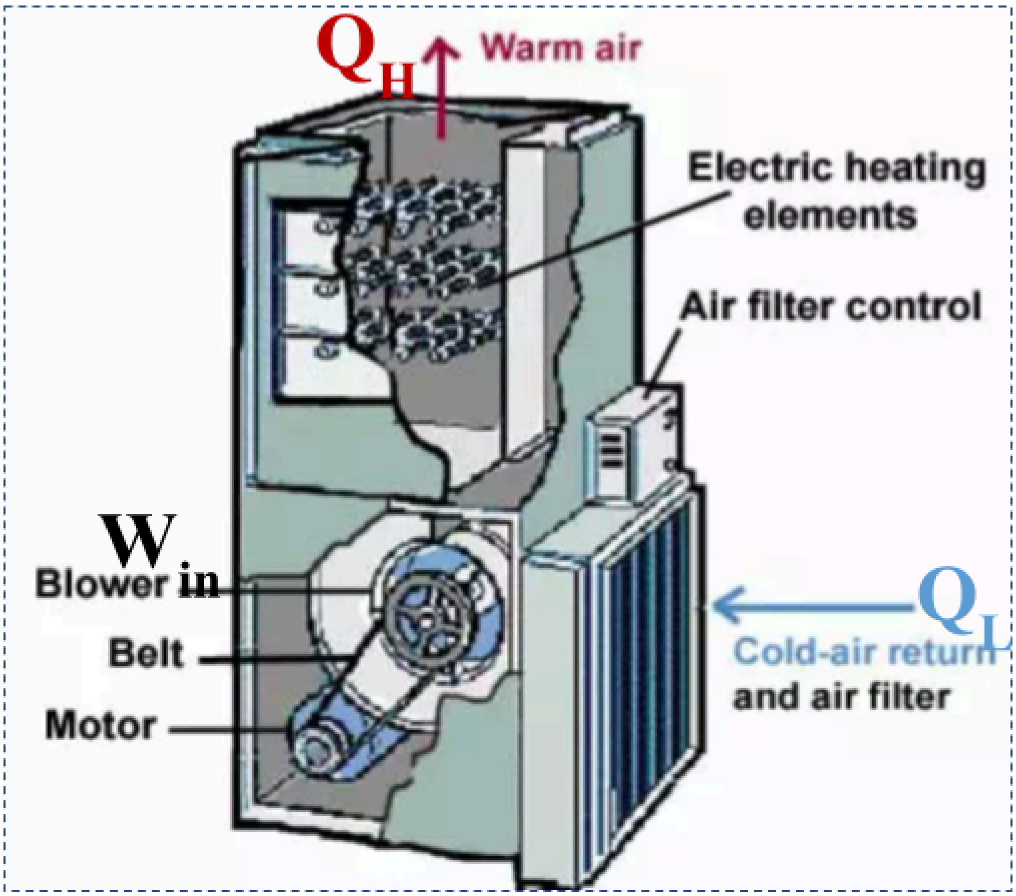

Figure 4: Electric Heater

(Vandervort, 2019)

$$\dot{E}_{in} - \dot{E}_{out} = 0$$

$$\dot{m}_{vapor,T_L} h_{vapor,T_L} + Q_H + \dot{W}_{in} = \dot{m}_{vapor,T_H} h_{vapor,T_H}$$

An electric heater starts by drawing in cold air from the indoors through an air filter. A motor drives the blower, which pushes the drawn-in air to the heat exchanger which, in this case, consists of electric heating elements. These electric heating elements heat up the air as it is pushed through them before exiting the heater and entering the room (Vandervort, 2019).

Design of the HVAC System

Table 1: Design

Devices	Quantity
Evaporative Cooler	1
Air Conditioner With a Heating Kit	1

Evaporative Cooler

4900 CFM Down-Draft Roof Evaporative Cooler for 1800 sq. ft. (Motor Not Included)

- 1800 sf cooling area
- 4900 CFM air volume

Air Conditioner

Direct Comfort 3.5 ton 16 seer single stage air conditioning system with electric heat

- 3.5 ton
- 16 seer
- 5 kW

Electric Heater

8 Kw Goodman Electric Strip Heat Kit with Circuit Breaker

- 8 kW
- 3 ton

Assumptions:

- Using the average humidity value for the specified month
- Using the average temperature for the specified month
- Assuming the family is using the basic SRP plan
- Assuming the efficiency of the device is included in the published specifications
- Because a water bill is complex, an average value of $31 for the use of 9000 gallons of water was used to determine how much it costs to use one gallon of water

Air Conditioning Calculation

Sample calculation for calculating the thermal energy in March:

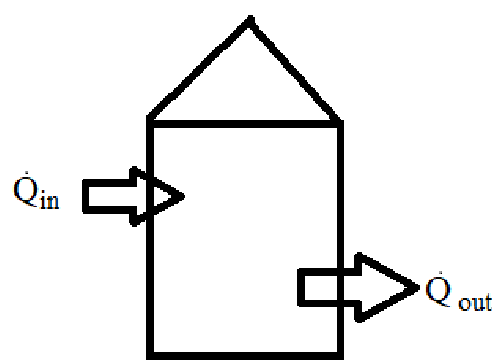

Figure 5: Heat transfer

Figure 5 shows that the amount of heat that is absorbed by the house must equal to the amount of heat that the air conditioner extracts from the house in order to keep it at a comfortable constant temperature.

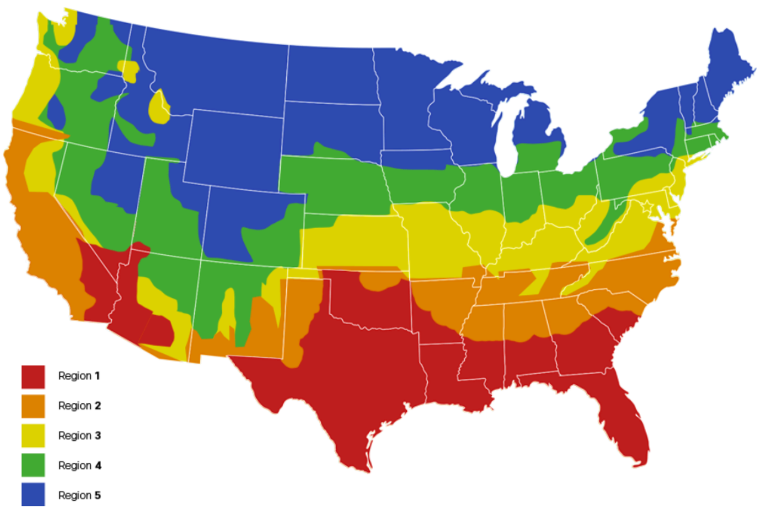

Figure 6: Selecting region based on the map (Air Conditioning Sizing Calculator, 2019)

AIR CONDITIONING SQUARE FOOTAGE RANGE BY CLIMATE ZONE

	ZONE 1	ZONE 2	ZONE 3	ZONE 4	ZONE 5
1.5 Tons	600 - 900 sf	600 - 950 sf	600 - 1000 sf	700 - 1050 sf	700 - 1100 sf
2 Tons	901-1200 sf	951 - 1250 sf	1001 - 1300 sf	1051 - 1350 sf	1101 - 1400 sf
2.5 Tons	1201 - 1500 sf	1251 - 1550 sf	1301 - 1600 sf	1351 - 1600 sf	1401 - 1650 sf
3 Tons	1501 - 1800 sf	1501 - 1850 sf	1601 - 1900 sf	1601 - 2000 sf	1651 - 2100 sf
3.5 Tons	1801 - 2100 sf	1851 - 2150 sf	1901 - 2200 sf	2001 - 2250 sf	2101 - 2300 sf
4 Tons	2101 - 2400 sf	2151 - 2500 sf	2201 - 2600 sf	2251 - 2700 sf	2301 - 2700 sf
5 Tons	2401 - 3000 sf	2501 - 3100 sf	2601 - 3200 sf	2751 - 3300 sf	2701 - 3300 sf

Figure 7: Air conditioner size (Air Conditioning Sizing Calculator, 2019)

Since the average single-family home size in Phoenix Arizona is 1832 square feet, the air conditioner size that is suitable for the home is 3.5 tons.

Knowing that one ton is equal to 12,000 $\frac{Btu}{hr}$, therefore,

$$(3.5 \text{ tons}) \left(\frac{12{,}000 \frac{Btu}{hr}}{1\ ton}\right) = 42{,}000 \frac{Btu}{hr}.$$

$42{,}000\ \frac{Btu}{hr}$ represents the cooling capacity for the air conditioner, which means that this is the maximum Btu's that the air conditioner can extract from the house in one hour. Therefore,

$$\dot{Q}_{in} = \dot{Q}_{out} = 42{,}000\ \frac{Btu}{hr}$$

However, the outside temperature is irregular and some days there will be more heat absorbed by the house than other days, which results in the need for the following calculation:

$$\frac{T_O - T_R}{T_{H,\ max} - T_R} = \frac{\dot{Q}_L}{42{,}000\ \frac{Btu}{hr}}$$

T_O: Average outside temperature for the month

T_R: The desired indoor house temperature

$T_{H,\ max}$: The highest average temperature of the hottest month

$\dot{Q}_L$: The rate of cooling load for the house

$$\dot{Q}_L = \left(\frac{T_O - T_R}{T_{H,\ max} - T_R}\right) * \left(42{,}000\ \frac{Btu}{h}\right)$$

The calculation above is a ratio that relates the medium to how much heat is extracted by the air conditioner. If the medium has a greater difference, then that means the air conditioner needs to extract more heat. The outside temperature (T_O) is the only number that will change through the calculations because the average temperature during the month is different for each month.

March:

T_O: 76.9 ˚F

T_R: 70 ˚F

$T_{H,\ max}$: 106.1 ˚F

$$\dot{Q}_L = \left(\frac{76.9\,^{\circ}F - 70\,^{\circ}F}{106.1\,^{\circ}F - 70\,^{\circ}F}\right) * \left(42,000\ \frac{Btu}{h}\right) = 8,027.70\ \frac{Btu}{h}$$

The SEER is seasonal energy efficiency rating and for the air conditioner unit that is chosen to use is:

SEER = 16 $\frac{Btu}{Wh}$

$$\text{SEER} = \frac{\dot{Q}_{in}}{\dot{w}_{E,\,in}} \rightarrow \dot{w}_{E,in} = \frac{\dot{Q}_{in}}{SEER}$$

$$\dot{w}_{E,in} = \frac{8{,}027.70\ \frac{Btu}{h}}{16\frac{Btu}{Wh}} = 501.73\ W$$

$$(250.83\text{ W})\ (4\text{ hours}) = 2006.93\text{ Wh} = 2.006\ \frac{kWh}{day}$$

$$(2.006\ \frac{kWh}{day})\ (\frac{31\ days}{1\ month}) = 62.21\text{ kWh}$$

$$(62.21\text{ kWh})\ (\frac{\$\ 0.0782}{1\ kWh}) = \$\ 4.87$$

This means that it cost $4.87 to run the air conditioner for four hours a day in the month of March.

Table 2: Months that Need Cooling

Month	Temperature (˚C)	$\dot{Q}_L$ ($\frac{Btu}{h}$)	Hours per day	Work (kWh)	Rate (USD per kWh)	Cost (USD)
March	76.9	8027.70	4	62.21	0.0782	4.87
August	104.4	40022.16	10	775.43	0.1157	89.72
September	99.80	34,670.36	8	520.06	0.1091	56.74
October	88.5	21,523.55	6	250.21	0.1091	27.30

The temperatures used for the heating load are the average highs for the specific months so the heating load for each month would be sufficient for the highest temperature.

Evaporative Cooler Calculations

Sample calculations for evaporative cooler uses the data and assumptions in July which is the hottest month in Arizona. All data presented in this paper is taken as the average values throughout a month using online sources presented on the reference page.

Assumptions

Table 3: July Data

Month	T_O (°F)	RH (%)	Hours of use	$\dot{Q}_L$ (Btu/hr)
July	106.1	31.60	15	42000

Where:

T_O : Outdoor temperature (°F)

RH : Relative humidity (%)

$\dot{Q}_L$: Cooling rate

The cooling capacity for July is 42,000 Btu/hr as the referred website suggests 3.5 tons for an 1832 ft^2 house. The website already assumes that the heat transfer rate occurs at the highest temperature in the summer in Phoenix region. Therefore, 42,000 Btu/hr is used as the average heat transfer rate for July.

Table 4: General Data

T_R (°F)	$T_{H, max}$ (°F)	ρ_{H_2O} (lbm/ft^3)	ρ_{air} (lbm/ft^3)	P_{atm} (psia)	M_{H20} (lbm/mol)	M_{air} (lbm/mol)	Cost per volume (\$/ft^3)	h_{fg} (Btu/lbm)	$P_{sat @ T_R}$ (psia)
70.0	106.1	62.2	0.0752	14.72	0.0397	0.0639	0.0258	1053.7	0.36334

Where:

T_R : Room temperature (°F)

$T_{H,\,max}$: Average outdoor temperature in the hottest month (°F)

ρ_{H_2O} : Density of water (lbm/ft^3)

ρ_{air} : Density of dry air (lbm/f^{t3})

P_{atm} : Atmospheric pressure in Phoenix (psia)

M_{H20} : Molar mass of water (lbm/mol)

M_{air} : Molar mass of dry air (lbm/mol)

h_{fg} : Latent heat of water (Btu/lbm)

$P_{sat\,@\,T_R}$: Saturation pressure at room temperature (psia)

Calculations for Water Cost per Month

$$\dot{m}_{H_2O}\, h_{fg} = \dot{Q}_L$$

Water mass flow rate: $\dot{m}_{H_2O} = \dot{Q}_L\, h_{fg} = \frac{\left(42{,}000\frac{Btu}{hr}\right)}{1053.7\frac{Btu}{lbm}} = 39.86\,\frac{lbm}{hr}$

Water volume flow rate: $\dot{V}_{H_2O} = \frac{\dot{m}_{H_2O}}{\rho_{H_2O}} = \frac{39.86\frac{lbm}{hr}}{62.2\frac{lbm}{ft^3}} = 0.641\,\frac{ft^3}{hr}$

Water cost per month:

$$C_{H_2O} = \frac{\$0.0258}{ft^3} \times 0.641\,\frac{ft^3}{hr} \times \frac{15\ hrs}{1\ day} \times \frac{31\ days}{1\ month} = \$7.43/month$$

Calculations for Evaporative Cooler Specifications (CFM)

Depending on the size of the house, the size of an evaporative cooler also changes accordingly. Therefore, it is important to see the maximum required air flow rate for the house to purchase a suitable unit.

Partial pressure at the outlet: $P_{H_2O} = X\ P_{atm}$

Where:

P_{H_2O} : Partial pressure at the respective relative humidity (psia)

X : Mole fraction

If the partial pressure at the outlet is 100% meaning the air holds the water molecules at 100% of its capacity, the partial pressure of water is very close to saturation pressure at room temperature (70°F). This means at 100% humidity, the partial pressure at the outlet would be:

$$P_{sat\,@\,T_R} = X\ P_{atm}$$

The researched ideal humidity of a room is in between 40% to 50% . To ensure the best health and comfort for the household, 40% will be used as the desired humidity level of the room. As the desired output humidity decreases, the partial pressure of water also decreases accordingly. Therefore, the ideal partial pressure at the outlet of the room can be calculated as follows.

$$\left(\frac{40\%-RH}{100\%-RH}\right) P_{sat\,@\,T_R} = X\ P_{atm}$$

Where RH is the relative humidity of the moist entering the evaporative cooler. According to table 3, July has a RH of 31.60% since it is sometimes the monsoon season.

$$\left(\frac{40\%-31.6\%}{100\%-31.6\%}\right) 0.36334\,psia = X\,14.72\,psia$$

$$X = \frac{\left(\frac{40\%-31.6\%}{100\%-31.6\%}\right) 0.36334\,psia}{14.72\,psia} = 0.0030313$$

Mole fraction at the outlet: $X = \frac{\dot{n}_{H_2O}}{\dot{n}_T} = \frac{\dot{n}_{H_2O}}{\dot{n}_{H_2O} + \dot{n}_{Air}} = \frac{1}{1+\frac{\dot{n}_{Air}}{\dot{n}_{H_2O}}} = \frac{1}{1+\frac{\frac{\dot{m}_{Air}}{M_{Air}}}{\frac{\dot{m}_{H_2O}}{M_{H_2O}}}}$

Air mass flow rate: $\dot{m}_{Air} = M_{Air}\frac{\dot{m}_{H_2O}}{M_{H_2O}}\left(\frac{1}{X} - 1\right)$

$$\dot{m}_{Air} = \left(0.0639\frac{lbm}{mol}\right)\left(\frac{39.86\frac{lbm}{hr}}{0.0397\frac{lbm}{mol}}\right)\left(\frac{1}{0.0030313} - 1\right) = 21100.64\frac{lbm}{hr}$$

Air volume flow rate: $\dot{V}_{Air} = \frac{\dot{m}_{Air}}{\rho_{Air}} = \frac{21100.64\frac{lbm}{hr}}{0.0752\frac{lbm}{ft^3}}$

$$280707.69\frac{ft^3}{hr}\frac{1hr}{60\ mins} = 4680\ \frac{ft^3}{min}$$

With similar calculations, the values of the other three months are shown in tables 5 and 6.

Table 5: Water Cost

Month	$\dot{Q}_L$ (Btu/hr)	$\dot{m}_{H_2O}$ (lbm/hr)	$\dot{V}_{H_2O}$ (ft^3/hr)	Cost of water per month
April	17684	16.78	0.270	$ 1.67
May	28853	27.38	0.440	$ 4.08
June	39440	37.43	0.602	$ 6.51
July	42000	39.86	0.641	$ 7.43
Total				$19.69

Where:

$\dot{m}_{H_2O}$: Water mass flow rate (lbm/hr)

$\dot{V}_{H_2O}$: Water volume flow rate (ft^3/hr)

The total cost of water from April to July is approximately $19.69.

Table 6: CFM

Month	T_O (°F)	RH (%)	Hours of use	P_{H_2O} (psia)	X	$\dot{m}_{Air}$ (lbm/hr)	$\dot{V}_{Air}$ (ft^3/hr)	CFM
April	85.2	27.8	8	0.0613954	0.0041709	6450	85801	1430
May	94.8	21.9	12	0.0842056	0.0057205	7661	101911	1699
June	103.9	19.4	14	0.0928636	0.0063087	9490	126243	2104
July	106.1	31.6	15	0.0446207	0.0030313	21101	280708	4678

The evaporative cooler is only used for four months of the summer since its efficiency decreases significantly when the humidity goes beyond 30%. The air volume flow rate in July is the maximum flow rate that is needed to cool the house in the hottest month of the year. As presented above, a CFM of 4680 is needed. The recommended model for a house 1832 ft^2 has a CFM of 4900. Since 4680 is smaller than 4900, the chosen unit is proven to be the suitable size for the house.

Electric Heater Calculation

Sample calculation for calculating the thermal energy in the month of December:

The idea for heating the house is the same as figure 5, the amount of heat the house is losing to the environment is equal to the amount of heat that must be supplied to the house. The source that is used to calculate the size of the air conditioner is also used to calculate the size of the heater for the house.

Based on the location and the size of the house the best suitable heater size is 3 tons. Therefore,

$$(3 \text{ tons}) \left(\frac{12{,}000 \frac{Btu}{hr}}{1\ ton}\right) = 36{,}000 \frac{Btu}{hr}.$$

$36{,}000 \frac{Btu}{hr}$ represents the heating capacity for the heater that is chosen and the maximum amount of heat that the heater could provide to the house in one hour.

$$\dot{Q}_{in} = \dot{Q}_{out} = 36,000 \frac{Btu}{hr}$$

$$\frac{T_O - T_R}{T_{L,\ min} - T_R} = \frac{\dot{Q}_H}{36{,}000 \frac{Btu}{hr}}$$

T_H: Average outside temperature for the month

T_L: The desired indoor house temperature

$T_{L,\ min}$: The lowest average temperature of the coldest month

$\dot{Q}_H$: The rate of heating load of the house

$$\dot{Q}_H = \left(\frac{T_H - T_L}{T_{H,\ max} - T_L}\right) * \left(54,960 \frac{Btu}{h}\right)$$

T_H: 44.8 ˚F

T_L: 70 ˚F

$T_{H,\ max}$: 44.8 ˚F

$$\dot{Q}_H = \left(\frac{44.8\,^{\circ}F - 70\,^{\circ}F}{44.8\,^{\circ}F - 70\,^{\circ}F}\right) * \left(36,000 \frac{Btu}{h}\right) = 36,000 \frac{Btu}{h}$$

SEER = 16 $\frac{Btu}{Wh}$

$$\text{SEER} = \frac{\dot{Q}_{in}}{\dot{w}_{E,\ in}} \rightarrow \dot{w}_{E,in} = \frac{\dot{Q}_{in}}{SEER}$$

$$\dot{w}_{E,in} = \frac{36{,}000 \frac{Btu}{h}}{16 \frac{Btu}{Wh}} = 2250\ W$$

$$(2250\text{ W})\ (4\text{ hours}) = 9000\text{ Wh} = 9.00\ \frac{kWh}{day}$$

$$(9.00\ \frac{kWh}{day})\ (\frac{31\ days}{1\ month}) = 279.0\text{ kWh}$$

$$(279.0\text{ kWh})\ (\frac{\$\ 0.0782}{1\ kWh}) = \$\ 21.82$$

This means that it cost $21.82 to run the heater for four hours a day in the month of December.

Table 7: Months that Need Heating

Month	Temperature (˚F)	$\dot{Q}_H\ (\frac{Btu}{h})$	Hours per day	Work (kWh)	Rate (USD per kWh)	Cost (USD)
December	44.8	36,000	4	297.00	0.0782	21.82
January	45.6	34,857.14	4	270.14	0.0782	21.13
February	48.7	30,428.57	4	220.61	0.0782	17.25
November	52.7	24,714.29	4	185.36	0.0782	14.49

The temperatures used for the heating load are the average lows for the specific months so the heating load for each month would be sufficient for the lowest temperature.

Table 8: Yearly Energy Cost with Evaporative Cooler

Month	Temperature (˚F)	Relative humidity (%)	Device	Hours per day	Cost ($ USD)
January	45.6	50.9	Heater	4	21.13
February	48.7	44.4	Heater	4	17.25
March	76.9	39.3	AC	4	4.87
April	85.2	27.8	Evaporative Cooler	8	4.56

May	94.8	21.9	Evaporative Cooler	12	9.85
June	103.9	19.4	Evaporative Cooler	14	11.12
July	106.1	31.6	Evaporative Cooler	16	13.93
August	104.4	36.2	AC	10	89.72
September	99.8	35.6	AC	8	56.74
October	88.5	36.9	AC	6	27.30
November	52.7	43.8	Heater	4	14.49
December	44.8	51.8	Heater	4	21.82
Water cost for the Evaporative Cooler					19.69
Total cost					312.45

Table 9: Yearly Energy Cost without Evaporative Cooler

Month	Temperature (˚F)	Relative humidity (%)	Device	Hours per day	Cost ($ USD)
January	45.6	50.9	Heater	4	21.13
February	48.7	44.4	Heater	4	17.25
March	76.9	39.3	AC	4	4.87
April	85.2	27.8	AC	8	20.74
May	94.8	21.9	AC	12	73.19
June	103.9	19.4	AC	14	112.95
July	106.1	31.6	AC	16	150.64
August	104.4	36.2	AC	10	84.50
September	99.8	35.6	AC	8	56.74

October	88.5	36.9	AC	6	27.30
November	52.7	43.8	Heater	4	14.49
December	44.8	51.8	Heater	4	21.82
Total cost					605.62

From November to February, the heater is used because the average temperature is lower than the desired temperature of 70°F. In March, August, September, and October, conventional air conditioner is used since the relative humidity for those months are over 32%, which is higher than the ideal humidity range for a swamp cooler. A humidity that is beyond 32% would require a bigger swamp cooler unit as its efficiency goes down significantly. From April to July, a swamp cooler is recommended to lower the electric bill. Since these months do not have a high humidity, a swamp cooler would be at its best efficiency. The total cost of the electric bill would be $293.16 lower per year if a swamp cooler is used for some of the hot months instead of using air conditioner for all of the hot months. If the priority is cost efficiency, then the plan with a swamp cooler is recommended. However, some consumers would rather pay the extra cost to have an air conditioner on for all the hot months for a few reasons. First off, in order to use a swamp cooler, windows and doors need to be open constantly. Secondly, swamp coolers increase the humidity of the air in the house. Lastly, an average swamp cooler costs about $600, considering the relative low amount of money one saves from the electric bill, it would take about two and a half years.until the consumer starts saving money. Nonetheless, in the long run, the consumer would end up saving about $3000 in ten years if a swamp cooler is implemented in their HVAC system.

Regulations

The design with a swamp cooler does not violate any state violation for there are no obstructions around the devices and that the sizes of the units are chosen for the appropriate area. Appropriate AC unit and evaporative cooler sizes were chosen for average home size, so the HVAC system will not be too large or too small for the home.

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